Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

beamer [faculty=wi]beamerthemefibeamer [utf8]inputenc [
main=polish, polish]babel
ragged2e booktabs tabularx tikz calc, shapes, backgrounds
amsmath, amssymb url listings

Presentation Title

Presentation Subtitle

Author's Name

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Outline for Section 1

Dark Frames

- Blind Text
- Structuring Elements
- Numerals and Mathematics
- Figures and Code Listings
- Citations and Bibliography

Light Frames

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Jabberwocky

Lewis Carroll

[overlay,remember picture] [anchor=south
east,xshift=-30pt,yshift=35pt] at (current page.south east) ;
'Twas brillig, and the slithy toves
Did gyre and gimble in the wabe;
All mimsy were the borogoves,
And the mome raths outgrabe.

"Beware the Jabberwock, my son!
The jaws that bite, the claws that catch!
Beware the Jubjub bird, and shun
The frumious Bandersnatch!"

Lists and locales

Lorem ipsum dolor sit amet

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Nechť již hříšné saxofony ďáblů rozzvučí síň úděsnými tóny waltzu, tanga a quickstepu! Nezvyčajné kŕdle šťastných figliarskych ďatľov učia pri kótovanom ústí Váhu mĺkveho koňa Waldemara obžierať väčšie kusy exkluzívnej kôry. The quick, brown fox jumps over a lazy dog. DJs flock by when MTV ax quiz prog. “Now fax quiz Jack!”

Text blocks

In plain, example, and alert flavour

This text is highlighted.

A plain block

This is a plain block containing some highlighted text.

An example block

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An alert block

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Definitions, theorems, and proofs

All integers divide zero

Definition

$\forall a, b \in Z : a \mid b \iff \exists c \in Z : a \cdot c = b$

Theorem

$\forall a \in Z : a \mid 0$

Proof

$\forall a \in Z : a \cdot 0 = 0$ $\square$

Numerals and Mathematics

Formulae, equations, and expressions

$$1234567890 \quad 1234567890 \quad \hat{x},\ \check{x},\ \tilde{a},\ \bar{a},\ \dot{y},\ \ddot{y} \quad \iint f(x,y,z)\,\mathrm{d}x\mathrm{d}y\mathrm{d}z$$

$$\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3+x}}} + \cfrac{1}{1+\frac{1}{2+\frac{1}{3+x}}} \qquad F : \begin{vmatrix} F''_{xx} & F''_{xy} & F'_x \\ F''_{yx} & F''_{yy} & F'_y \\ F'_x & F'_y & 0 \end{vmatrix} = 0$$

$$\iint\limits_{x \in R^2} \langle \mathrm{x}, \mathrm{y} \rangle \,\mathrm{dx} \qquad \overline{\overline{a\alpha^2} + \underline{b\beta} + \overline{\overline{d\delta}}} \qquad]0,1[+ \lceil x \rfloor - \langle x, y \rangle$$

$$e^x \approx 1 + x + x^2/2! + + x^3/3! + x^4/4! \qquad \binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$

Figures

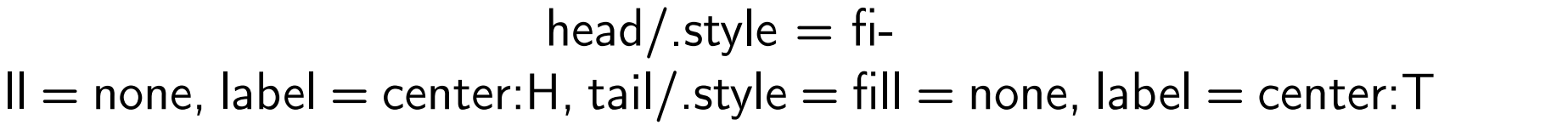
Tables, graphs, and images

Xrrr **Faculty**	**With TEX**	**Total**	**%**
Faculty of Informatics	1 716	2 904	59.09
Faculty of Science	786	5 275	14.90
Faculty of Economics and Administration	64	4 591	1.39
Faculty of Arts	69	10 000	0.69
Faculty of Medicine	8	2 014	0.40
Faculty of Law	15	4 824	0.31
Faculty of Education	19	8 219	0.23
Faculty of Social Studies	12	5 599	0.21
Faculty of Sports Studies	3	2 062	0.15

Tabel: The distribution of theses written using TEX during 2010–15 at MU

Figures

Tables, graphs, and images



Gambar: Tree of probabilities – Flipping a coin[1]

[1] A derivative of a diagram from `texample.net` by cis, CC BY 2.5 licensed

Code listings

An example source code in C

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Citations

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Bibliography

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Outline for Section 2

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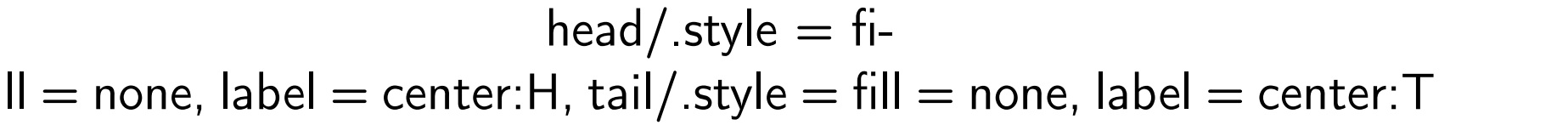
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